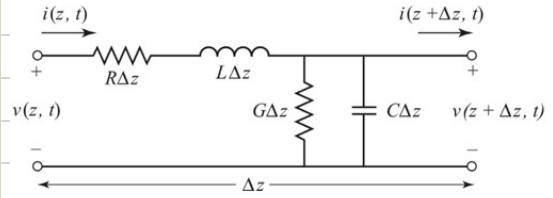
TLWA Formulas & Derivations

Hitesh Pranav

- – Derivation
- – Formula

UNIT-1

Equation of Voltage and Current in transmission line



→ Apply KVL to circuit,

$$v(z,t) - i(z,t).R\Delta z - \frac{di(z,t)}{dt}.L\Delta z - v(z+\Delta z,t) = 0$$

$$v(z+\Delta z,t) - v(z,t) = -\Delta z\left(Ri(z,t) + L\frac{\partial i}{\partial t}(z,t)\right)$$

$$\frac{v(z+\Delta z,t) - v(z,t)}{\Delta z} = -\left(Ri(z,t) + L\frac{\partial i}{\partial t}(z,t)\right)$$

Δz is very small, so apply $\lim_{\Delta z \to 0}$

$$\lim_{\Delta z\to 0}\frac{v(z+\Delta z,t) - v(z,t)}{\Delta z} = -\left(Ri(z,t) + L\frac{\partial i}{\partial t}(z,t)\right)$$

$$\frac{\partial v(z,t)}{\partial z} = -Ri(z,t) - L\frac{\partial i(z,t)}{\partial t}$$

Convert to Frequency domain

$$\frac{\partial v(z,t)}{\partial z} = -(R+j\omega L)\,i(z,t) \longrightarrow (1)$$

→ Apply KCL to circuit,

$$i(z,t) = G\Delta z\, v(z,t) + C\Delta z\frac{\partial v(z,t)}{\partial t} + i(z+\Delta z,t)$$

$$\frac{i(z+\Delta z,t) - i(z,t)}{\Delta z} = -\left(Gv(z,t) + C\frac{\partial v(z,t)}{\partial t}\right)$$

Δz is very small, so apply $\lim_{\Delta z \to 0}$

$$\lim_{\Delta z\to 0}\frac{i(z+\Delta z,t) - i(z,t)}{\Delta z} = -\left(Gv(z,t) + C\frac{\partial v(z,t)}{\partial t}\right)$$

$$\frac{\partial i(z,t)}{\partial z} = -\left(Gv(z,t) + C\frac{\partial v(z,t)}{\partial t}\right)$$

Convert to Frequency domain

$$\frac{\partial i(z,t)}{\partial z} = -(G+j\omega C)\,v(z,t) \longrightarrow (2)$$

→ Apply sinusoidal steady state equation to (1) & (2)

$$\frac{dv(z)}{dz} = -(R+j\omega L)\,I(z) \longrightarrow (3)$$

$$\frac{dI(z)}{dz} = -(G+j\omega C)\,v(z) \longrightarrow (4)$$

→ Differentiate (3) wrt z

$$\frac{d^2v(z)}{dz^2} = -(R+j\omega L)\frac{dI}{dz}(z)$$

$$= -(R+j\omega L).-(G+j\omega C).v(z)$$

$$= (R+j\omega L)(G+j\omega C)\,v(z)$$

Assume $(R+j\omega L)(G+j\omega C) = \gamma^2$

$$\frac{d^2v(z)}{dz^2} = \gamma^2 v(z) \Rightarrow \frac{d^2v(z)}{dz^2} - \gamma^2 v(z) = 0 \longrightarrow (5)$$

Similarly, $\frac{d^2I(z)}{dz^2} - \gamma^2 I(z) = 0 \longrightarrow (6)$

Upon Solving (5) & (6)

$$v(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \longrightarrow (7)$$

↳ This is voltage eqⁿ of Tx Line

$$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \longrightarrow (8)$$

↳ This is current eqⁿ of Tx Line

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)} = \alpha + j\beta$$

where γ: Complex Propagation Constant

α: Attenuation Constant

β: Phase constant

From (3), $I(z) = \frac{-1}{R+j\omega L}.\frac{dV(z)}{dz}$

$$= \frac{-1}{R+j\omega L}.\frac{d\left(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}\right)}{dz}$$

$$= \frac{-1}{R+j\omega L}.\left(-V_0^+\gamma e^{-\gamma z} + V_0^-\gamma e^{\gamma z}\right)$$

$$I(z) = \frac{\gamma}{R+j\omega L}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right) \longrightarrow (9)$$

Comparing with (8) & (9)

$$\frac{V_0^+}{I_0^+} = \frac{-V_0^-}{I_0^-} = \frac{R+j\omega L}{\gamma} = \frac{R+j\omega L}{\sqrt{(R+j\omega L)(G+j\omega C)}} = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = z_0$$

Where, z_0 : Characteristic Impedance

Converting back to time domain

$$v(z,t) = V_0^+\cos(\omega t - \beta z + \phi^+)\,e^{-\alpha z} + V_0^-\cos(\omega t - \beta z + \phi^-)\,e^{\alpha z}$$

where, ϕ^x : Phase angle of Complex Voltage V_0^x

Wavelength of line, $\lambda = \frac{2\pi}{\beta}$

Phase Velocity, $v_p = \frac{\omega}{\beta}$

→ For Lossless line

Set $R = G = 0$ in γ

$\gamma = \alpha + j\beta = j\omega\sqrt{LC} \Rightarrow \beta = \omega\sqrt{LC}$

$\alpha = 0$

So attenuation constant $\alpha = 0$ for lossless line

and $Z_0 = \sqrt{\frac{L}{C}}$

Then general solutions for voltage & current on lossless transmission line are

$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}$

$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}$

Then wavelength, $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{LC}}$

phase velocity, $v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$

Some imp formulas,

$I_S = \frac{V_S}{Z_0}$

$P_S = I_S \cdot V_S$

$I_R = I_S \cdot e^{-\gamma l}$

$V_R = I_R \cdot Z_L$

$P_R = V_R \cdot I_R \cdot \cos\theta$

$\eta = \frac{P_R}{P_S}$

Also for calculations,

$e^{j\theta} = \cos\theta + j\sin\theta$

$e^{-j\theta} = \cos\theta - j\sin\theta$

$\cos\theta = (e^{j\theta} + e^{-j\theta})/2$

$\sin\theta = (e^{j\theta} - e^{-j\theta})/2$

$(r\angle\theta)^{y/n} = r^{y/n}\angle\theta/n$

→ For terminated lossless Tx line (No Reflection, Max Power)

$Z_L = Z_0$

total voltage = sum of incident & reflected voltages

$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}$

$I(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}$

The total voltage & current at the load are related by the load imp, so at $z = 0$

$Z_L = \frac{V_L}{I_L} = \frac{V(0)}{I(0)} = \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} Z_0$

Then, $V_0^- = \frac{Z_L - Z_0}{Z_L + Z_0} V_0^+$

Reflection Coefficient, $\Gamma(0) = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\text{amplitude of reflected voltage}}{\text{amplitude of incident voltage}}$ at $l = 0 / z = 0$

$\Gamma(l) = \frac{V_0^- e^{-j\beta l}}{V_0^+ e^{j\beta l}} = \frac{V_0^-}{V_0^+} e^{-2j\beta l} = \Gamma(0) e^{-2j\beta l}$

→ For Standing Waves,

The voltage & current on the line consist of a superposition of incident & reflected wave and such waves

$V(z) = V_0^+ e^{j\beta z}\left[1 + \Gamma e^{2j\beta z}\right]$

$I(z) = \frac{V_0^+ e^{-j\beta z}}{Z_0}\left[1 - \Gamma e^{2j\beta z}\right]$

at $z = -l$ and $\Gamma = |\Gamma| e^{j\theta}$

$V(z) = |V_0^+|\left(1 + |\Gamma| e^{j(\theta - 2\beta l)}\right) \longrightarrow V_{max} = |V_0^+|(1 + |\Gamma|)$

$V_{min} = |V_0^+|(1 - |\Gamma|)$

Standing Wave Ratio, $SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$ $\left(1 \le SWR < \infty,\ SWR = 1 \text{ for matched load}\right)$

Distance b/w 2 successive maxima & minima, $l = \frac{2\pi}{2\beta} = \frac{2\pi}{2\left(\frac{2\pi}{\lambda}\right)} = \frac{\lambda}{2}$

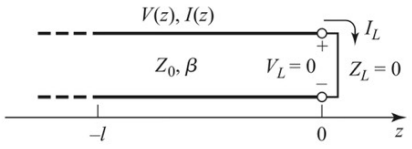
Distance b/w 2 successive maxima & minima, $l = \frac{\pi}{2\beta} = \frac{\pi}{2\left(\frac{2\pi}{\lambda}\right)} = \frac{\lambda}{4}$

→ Input Impedance, $z = -l$

$$Z_{in} = \frac{V(-l)}{I(-l)} = Z_0\left(\frac{1+\Gamma e^{-j2\beta l}}{1-\Gamma e^{-j2\beta l}}\right)$$

$$Z_{in} = Z_0\left(\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right)$$

→ Short circuited line, $Z_{in} = jZ_0\tan\beta l = Z_{sc}$ $(Z_L = 0)$



$\Gamma = -1$

VSWR = ∞ (Voltage SWR)

$V(z) = V_0^+\left(e^{-j\beta z} - e^{j\beta z}\right) = -2j\,V_0^+\sin\beta z$

$I(z) = \frac{V_0^+}{Z_0}\left(e^{-j\beta z} + e^{j\beta z}\right) = \frac{2V_0^+}{Z_0}\cos\beta z$

Now,

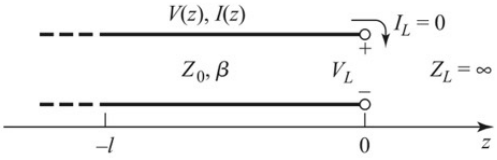
at $l = 0$, $\beta l = \frac{2\pi}{\lambda}\times 0 = 0$

at $l = \lambda/4$, $\beta l = \frac{2\pi}{\lambda}\times\frac{\lambda}{4} = \pi/2$

at $l = \lambda/2$, $\beta l = \frac{2\pi}{\lambda}\times\frac{\lambda}{2} = \pi$

at $l = 3\lambda/2$, $\beta l = \frac{2\pi}{\lambda}\times\frac{3\lambda}{2} = 3\pi$

→ Open circuited line, $Z_{in} = -jZ_0\cot\beta l = Z_{oc}$ $(Z_L = \infty)$



$\Gamma = 1$

VSWR = 0 (Voltage SWR)

→ Perfectly Matched Load, $Z_{in} = Z_0$ $(Z_L = Z_0)$

$\Gamma = 0$

VSWR = 1

→ Half Waveline, $l = \lambda/2 \implies \beta l = \pi$

$$Z_{in} = Z_0\frac{(Z_L + jZ_0\tan\beta l)}{(Z_0 + jZ_L\tan\beta l)} = Z_0\cdot\frac{Z_L}{Z_0} = Z_L \qquad (\tan\beta l = \tan\pi = 0)$$

→ $R_{max} = VSWR\cdot Z_0$; $R_{min} = \frac{Z_0}{VSWR}$

→ Average Power Flow in Tx line,

$V(z) = V_0^+\left(e^{-j\beta z} + \Gamma e^{j\beta z}\right)$ ⟶ ①

$I(z) = I_0^+\left(e^{-j\beta z} - \Gamma e^{j\beta z}\right)$ ⟶ ②

According to Poynting Theorem,

$$P_{avg} = \frac{1}{2}[E\times H^*]$$

$$= \frac{1}{2}\,Re\left[V(z).I(z)^*\right]$$

$$= \frac{1}{2}\frac{|V_0^+|^2}{Z_0}\,Re\left\{1 - \Gamma^* e^{-j2\beta z} + \Gamma e^{j2\beta z} - \Gamma\Gamma^*\right\} \qquad \left(I(z)^* = \frac{V_0^+}{Z_0}\left(e^{j\beta z} - \Gamma^* e^{-j\beta z}\right)\right)$$

$$P_{avg} = \frac{1}{2}\frac{|V_0^+|^2}{Z_0}\left[1 - |\Gamma|^2\right]$$

$\left(\Gamma e^{j2\beta z} = A,\ \Gamma^* e^{-j2\beta z} = A^*\right) \rightarrow Re(A - A^*) = Re(2j\,Im(A)) = 0$

For $\Gamma = 0$, $P_{avg} = \frac{|V_0^+|^2}{2Z_0}$

For $\Gamma = 1$, $P_{avg} = 0$

→ Return Loss,

Loss when load is mismatched ($z_L \neq z_0$) & all available power from generator is delivered to load

$RL = -20 \log |\Gamma|$ dB

→ Transmission Coefficient,

$T = 1 + \Gamma = \frac{2 z_L}{z_L + z_0}$

→ Insertion Loss,

The transmission coefficient between 2 points in a circuit

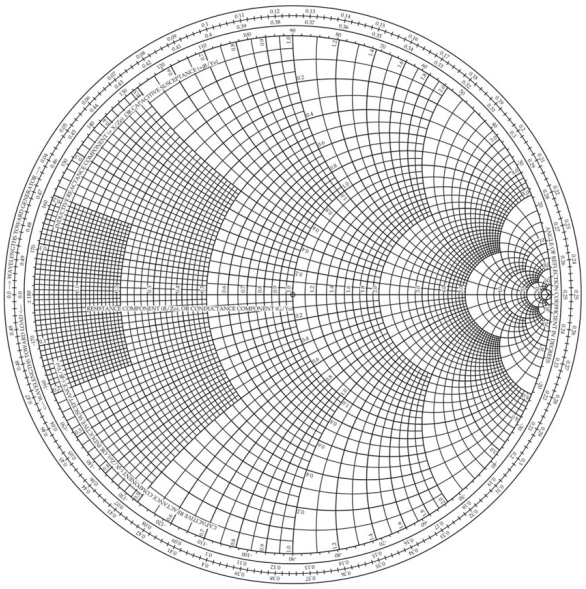
$IL = -20 \log |T|$ dB

→ Smith Chart → Horizontal Axis - Resistance

Vertical Axis - Reactance

$\Gamma = |\Gamma| e^{j\theta} \longrightarrow -180° \leq \theta \leq 180°$

↳ Radius, $|\Gamma| \leq 1$



→ Only deals with normalized values

$z = \frac{Z}{Z_0} \longrightarrow$ Normalized impedance

→ $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\frac{Z_L}{Z_0} - \frac{Z_0}{Z_0}}{\frac{Z_L}{Z_0} + \frac{Z_L}{Z_0}} = \frac{z_L - 1}{z_L + 1} = |\Gamma| e^{j\theta}$

$z_L = \frac{1 + |\Gamma| e^{j\theta}}{1 - |\Gamma| e^{j\theta}}$

Let $\Gamma = \Gamma_r + j\Gamma_i$ & $z_L = r_L + jx_L$

$z_L = r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}$

$= \frac{[(1 + \Gamma_r) + j\Gamma_i]}{[(1 - \Gamma_r) - j\Gamma_i]} \cdot \frac{(1 - \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) + j\Gamma_i}$

$= \frac{1 - \Gamma_r^2 - \Gamma_i^2 + 2j\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$

$r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} \longrightarrow ①$ & $x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2} \longrightarrow ②$

Rearranging ①, $\left(\Gamma_r - \frac{r_L}{1 + r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L}\right)^2 \Rightarrow$ Constant Res

(Circle equation)

So centre $= \left(\frac{r_L}{1 + r_L}, 0\right)$ radius $= \frac{1}{1 + r_L}$

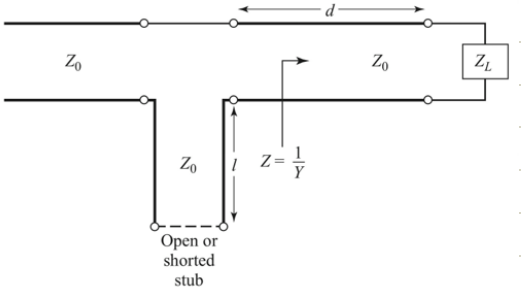
Rearranging ②, $(\Gamma_r - 1)^2 + \left(\Gamma_i^2 - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2$

So centre $= \left(1, \frac{1}{x_L}\right)$ radius $= \frac{1}{x_L}$

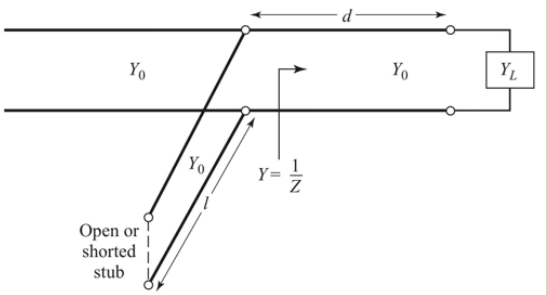
Total length of smith chart $= \frac{\lambda}{2} = 0.5\lambda$

→ Stub - Section of Tx line used for impedance matching

- Either Short Circuit / Open Circuit , Series & Shunt Stub Matching
  - → Stub in parallel with load
  - → Stub in series with load
- Less Radiation Loss ⇒ S.C Shunt



Series , Z chart , 1+jx



Shunt , Y chart , 1+jb

→ No reflection should be seen by generator

→ Scattering parameters/matrices are essential to analyze microwave devices because :

i) Active devices operates at constant frequencies & it gives instability as frequency changes

ii) z & y parameters are impractical at high frequencies and hard to realize open & short circuit above 1GHz

iii) The measurement of instantaneous voltage & current lose its physical meaning because of its distributed effects



$$S_{11} = \frac{b_1}{a_1}\Big|_{a_2=0} \qquad S_{21} = \frac{b_2}{a_1}\Big|_{a_2=0}$$

$$S_{12} = \frac{b_1}{a_2}\Big|_{a_1=0} \qquad S_{22} = \frac{b_2}{a_2}\Big|_{a_1=0}$$

→ $V = V_0^+ + V_0^-$ & $I = \frac{V_0^+ - V_0^-}{Z_0}$ ⇒ $V_0^+ = \frac{V + IZ_0}{2}$ & $V_0^- = \frac{V - IZ_0}{2}$

Avg power , $P_L = \frac{1}{2} Re\{V I^*\} = \frac{1}{2Z_0}\{|V_0^+|^2 - |V_0^-|^2\}$

Normalized incident vector $a = \frac{V_0^+}{\sqrt{Z_0}} = \frac{V_0 + IZ_0}{2\sqrt{Z_0}} = I_0^+ Z_0$

Normalized reflected vector $b = \frac{V_0^-}{\sqrt{Z_0}} = \frac{V_0 - IZ_0}{2\sqrt{Z_0}} = I_0^- Z_0$

Incident Power $P^+ = |a|^2$

Reflected Power $P^+ = |b|^2$

Reflection Coeff. $\Gamma = b/a$

Actual Power delivered to load , $P = |a|^2 - |b|^2$

$$VSWR = \frac{|a| + |b|}{|a| - |b|}$$

For n-port n/w, $a_i = \frac{1}{2}\left(\frac{V_i + Z_0 I_i}{\sqrt{Z_0}}\right)$ & $b_i = \frac{1}{2}\left(\frac{V_i - Z_0 I_i}{\sqrt{Z_0}}\right)$

→ Properties :

i) s matrix is a square matrix of nxn where n is no. of ports of a given device

$$[S] = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1n} \\ S_{21} & S_{22} & \cdots & S_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ S_{n1} & S_{n2} & \cdots & S_{nn} \end{bmatrix}$$

ii) Identity property : $[S][S]^* = I$

iii) Symmetric property : $S_{ij} = S_{ji}$ (Only for symmetric ports)

iv) Reciprocal matrix : $[S] = [S]^T$

v) Lossless network : $[S]^T[S]^* = I$

$|S_{11}|^2 + |S_{12}|^2 = 1$ (For 2x2) } Row

$|S_{11}|^2 + |S_{12}|^2 + \cdots + |S_{1n}|^2 = 1$ (For nxn) } Row

$|S_{11}|^2 + |S_{21}|^2 = 1$ (For 2x2) } Column

$|S_{11}|^2 + |S_{21}|^2 + \cdots + |S_{n1}|^2 = 1$ (For nxn) } Column

UNIT-2

→ Waveguides are complete hollow metallic structure through which E & M Fields are transmitted

→ Tx lines use TEM ; Waveguides use only TE or TM mode, no TEM . They propagate in z-direction

→ Waveguides propagate EM waves due to reflection of inner walls

→ Impedance of Tx usually 50Ω
Impedance of waveguide depends on mode

→ In waveguides, there is cutoff freq into different modes of propagation with TE & TM mode
In Tx lines , there isn't cutoff freq for TEM mode

→ In waveguides, lower attenuation

Waveguides !
- → Lower losses
- → Higher Power handling
- → Reduced Power Dispersion
- → No Radiation Losses

→ In far field of free space, E & M waves are $\perp^{ar}$ to each other & normal to direction of propagation of wave.
This is TEM wave
If only transverse E wave exists, then it is TE mode wave (No component of electric wave in direction of prop.)
If only transverse M wave exists, then it is TM mode wave (No component of magnetic wave in direction of prop.)

→ E & M wave eq^n's derived from Maxwell Eq^n's in time domain

$\nabla \times E = -\frac{\partial B}{\partial t}$ ⟶ (1) Potential

$\nabla \times H = J + \frac{\partial D}{\partial t}$ ⟶ (2) Ampere Circuit Law

$\nabla . D = \rho_v$ ⟶ (3) Gauss Electric Field

$\nabla . B = 0$ ⟶ (4) Gauss Magnetic Field

$D = \varepsilon E$
$B = \mu H$
$J_c = \sigma E$
$\varepsilon = \varepsilon_r \varepsilon_0$
$\mu = \mu_r \mu_0$

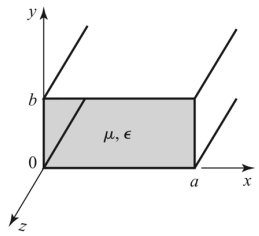
If sinusoidal f^n is assumed in the form e^{jwt}, then $\frac{\partial}{\partial t}$ is replaced with jw

$\nabla \times E = -jw\mu H$ ⟶ (5)

$\nabla \times H = jw \varepsilon E$ ⟶ (6) $\left(J_c = \sigma E = D \right)$

$\nabla . D = \rho_v$ ⟶ (7)

$\nabla . B = 0$ ⟶ (8)



In free space, space charge density is 0 & in a perfect conductor, time varying static field doesn't exist

$\nabla . D = 0$ ⟶ (9)

$\nabla . E = 0$ ⟶ (10)

Now take curl on (5)

$\nabla \times \nabla \times E = -jw\mu (\nabla \times H)$

$-\nabla^2 E + \nabla(\cancel{\rho}^{0} . E) = -jw\mu (jw \varepsilon E)$

$\nabla^2 E = j^2 w^2 \mu \varepsilon E$

$\nabla^2 E = \gamma^2 E$ where $\gamma = \sqrt{j^2 w^2 \mu \varepsilon} = jw\sqrt{\mu \varepsilon}$

Similarly,
$\nabla^2 H = \gamma^2 H$

$\gamma = \alpha + j\beta$
↳ Intrinsic Propagation Constant of Medium

→ When waves travel longitudinally down the guide, plane waves are reflected from the wall
This results in either E or M field in direction of propagation of resultant wave . No longer TEM wave

λ ⟶ λ_n (normal to reflecting surface)
↳ λ_p (parallel to reflecting surface)

In lossless waveguide, modes maybe classified as only 2 modes
↳ transverse electric (TE)
↳ transverse magnetic (TM)

In rectangular waveguide, TE_{mn} or TM_{mn}
where, m : no of half waves of E or M intensity in x direction
n : no of half waves in y direction if propagation assumed in z direction

Solutions of Wave Eq^n's in Rectangular Coordinates

→ Vector wave equations

$\nabla^2 E = \gamma^2 E$

$\nabla^2 H = \gamma^2 H$

$\gamma = j\omega\sqrt{\mu\varepsilon}$

Helmholtz Equation $\Rightarrow \nabla^2\psi = \gamma^2\psi$

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = \gamma^2\psi \longrightarrow (3)$$

$\left(\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$

Let $\psi = X(x).Y(y).Z(z) \longrightarrow (4)$

Substitute (4) in (3),

$$\gamma^2 = \frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} \longrightarrow (5)$$

Let $\frac{1}{X}\frac{d^2X}{dx^2} = -K_x^2$, $\frac{1}{Y}\frac{d^2Y}{dy^2} = -K_y^2$, $\frac{1}{Z}\frac{d^2Z}{dz^2} = -K_z^2$

Then , $-K_x^2 - K_y^2 - K_z^2 = \gamma^2 \longrightarrow (6)$

$$\frac{d^2X}{dx^2} = -K_x^2 X \quad , \quad \frac{d^2Y}{dy^2} = -K_y^2 Y \quad , \quad \frac{d^2Z}{dz^2} = -K_z^2 Z$$

General Solution

$X = A\sin(K_x x) + B\cos(K_x x)$

$Y = C\sin(K_y y) + D\cos(K_y y)$

$Z = E\sin(K_z z) + F\cos(K_z z)$

$\psi = (A\sin(K_x x) + B\cos(K_x x))(C\sin(K_y y) + D\cos(K_y y))(E\sin(K_z z) + F\cos(K_z z))$

The direction of propagation of wave is assumed to be $+z$, So the propagation constant γ_g in the guide differs from the intrinsic propagation γ

Let $\gamma_g^2 = \gamma^2 + K_x^2 + K_y^2$

$= \gamma^2 + K_c^2$ where K_c: cut-off wave number

$\gamma_g^2 = -\omega^2\mu\varepsilon + K_c^2$ (or) $K_c^2 = \gamma_g^2 + \omega^2\mu\varepsilon$

Case (i) : No propagation in the waveguide if $K_c^2 = \omega^2\mu\varepsilon$ therefore $\gamma_g = 0$

This is a critical condition for cut-off propagation

cutoff frequency , $f = \frac{1}{2\pi}\cdot\frac{K_c}{\sqrt{\mu\varepsilon}}$ $\Rightarrow f = f_c$

Case (ii) : Wave propagating in waveguide if $K_c^2 < \omega^2\mu\varepsilon$ therefore $\gamma_g = \pm j\beta_g$

$K = \frac{2\pi f}{v} = \sqrt{\mu\varepsilon}\,2\pi f$

$K_c = \frac{2\pi f_c}{v} = \sqrt{\mu\varepsilon}\,2\pi f_c$

$\beta = \sqrt{K^2 - K_c^2} = 2\pi f\sqrt{\mu\varepsilon}\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$ $\Rightarrow f > f_c$

Case (iii) : Wave will be attenuated if $K_c^2 > \omega^2\mu\varepsilon$ therefore $\gamma_g = \pm\alpha_g$

$\gamma_g = \omega\sqrt{\mu\varepsilon}\sqrt{\left(\frac{f_c}{f}\right)^2 - 1}$ $\Rightarrow f < f_c$

Solution of Helmholtz eq^n will be $\psi = [A\sin(K_x x) + B\cos(K_x x)][C\sin(K_y y) + D\cos(K_y y)]\cdot e^{-j\beta_g z}$

→ TE Modes

Dimensions $0 \le x \le a$ & $0 \le y \le b$

Propagation along z-axis

Perfect Electric Conductor (PEC) walls

Filled with homogeneous medium

TE : Transverse Electric

$E_z = 0$ & $H_z \neq 0$

So, $\nabla^2 H_z = \gamma^2 H_z$ ⟶ Helmholtz Equation

$\Psi = [A\sin(k_x x) + B\cos(k_x x)][C\sin(k_y y) + D\cos(k_y \cdot y)] \cdot e^{-j\beta_g z}$ ⟶ General Solution

Applying Boundary Conditions, $E_x = 0$ @ $y = 0$ & $y = b$ $\Rightarrow k_y = \frac{n\pi}{b}$

$E_y = 0$ @ $x = 0$ & $x = a$ $\Rightarrow k_x = \frac{m\pi}{a}$

So, $H_z = A_{mn} \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$ (Other formulas in summary table!)

→ Degenrate Modes,

If $f_{c_{m_1,n_1}} = f_{c_{m_2,n_2}}$ (2 different modes with same cutoff frequency)

Rectangular guides ordinarily have dimensions $a = 2b$ ratio

→ Dominant Mode

Mode with lowest cut-off frequency in a particular guide

$\lambda_{c_{min}} = \frac{2ab}{\sqrt{m^2b^2 + n^2a^2}}$

↳ TE Mode ⇒ TE_{00} doesn't exist because $f_c = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{0}{a}\right)^2 + \left(\frac{0}{b}\right)^2} = 0$ ✗

$TE_{01} \Rightarrow f_c = \frac{1}{2b\sqrt{\mu\varepsilon}}$

$TE_{10} \Rightarrow f_c = \frac{1}{2a\sqrt{\mu\varepsilon}}$

↳ but $a > b$ so TE_{10} has lowest cut off frequency

→ TM Modes

Dimensions $0 \le x \le a$ & $0 \le y \le b$

Propagation along z-axis

Perfect Electric Conductor (PEC) walls

Filled with homogeneous medium

TE : Transverse Magnetic

$E_z \neq 0$ & $H_z = 0$

So, $\nabla^2 E_z = \gamma^2 E_z$ ⟶ Helmholtz Equation

$\Psi = [A\sin(k_x x) + B\cos(k_x x)][C\sin(k_y y) + D\cos(k_y \cdot y)] \cdot e^{-j\beta_g z}$ ⟶ General Solution

Applying Boundary Conditions, $E_z = 0$ @ $y = 0$ & $y = b$ $\Rightarrow k_y = \frac{n\pi}{b}$

$E_z = 0$ @ $x = 0$ & $x = a$ $\Rightarrow k_x = \frac{m\pi}{a}$

So, $E_z = B_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}$

Imp : For sin function to exist, $m \neq 0$ & $n \neq 0$ because if they were 0, $E_z = 0$ everywhere

Lowest TM mode : TM_{11}

Quantity	TE_{mn} Mode	TM_{mn} Mode
k	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$
k_c	$\sqrt{(m\pi/a)^2+(n\pi/b)^2}$	$\sqrt{(m\pi/a)^2+(n\pi/b)^2}$
β	$\sqrt{k^2-k_c^2}$	$\sqrt{k^2-k_c^2}$
λ_c	$\dfrac{2\pi}{k_c}$	$\dfrac{2\pi}{k_c}$
λ_g	$\dfrac{2\pi}{\beta}$	$\dfrac{2\pi}{\beta}$
v_p	$\dfrac{\omega}{\beta}$	$\dfrac{\omega}{\beta}$
α_d	$\dfrac{k^2\tan\delta}{2\beta}$	$\dfrac{k^2\tan\delta}{2\beta}$
E_z	0	$B\sin\dfrac{m\pi x}{a}\sin\dfrac{n\pi y}{b}e^{-j\beta z}$
H_z	$A\cos\dfrac{m\pi x}{a}\cos\dfrac{n\pi y}{b}e^{-j\beta z}$	0
E_x	$\dfrac{j\omega\mu n\pi}{k_c^2 b}A\cos\dfrac{m\pi x}{a}\sin\dfrac{n\pi y}{b}e^{-j\beta z}$	$\dfrac{-j\beta m\pi}{k_c^2 a}B\cos\dfrac{m\pi x}{a}\sin\dfrac{n\pi y}{b}e^{-j\beta z}$
E_y	$\dfrac{-j\omega\mu m\pi}{k_c^2 a}A\sin\dfrac{m\pi x}{a}\cos\dfrac{n\pi y}{b}e^{-j\beta z}$	$\dfrac{-j\beta n\pi}{k_c^2 b}B\sin\dfrac{m\pi x}{a}\cos\dfrac{n\pi y}{b}e^{-j\beta z}$
H_x	$\dfrac{j\beta m\pi}{k_c^2 a}A\sin\dfrac{m\pi x}{a}\cos\dfrac{n\pi y}{b}e^{-j\beta z}$	$\dfrac{j\omega\epsilon n\pi}{k_c^2 b}B\sin\dfrac{m\pi x}{a}\cos\dfrac{n\pi y}{b}e^{-j\beta z}$
H_y	$\dfrac{j\beta n\pi}{k_c^2 b}A\cos\dfrac{m\pi x}{a}\sin\dfrac{n\pi y}{b}e^{-j\beta z}$	$\dfrac{-j\omega\epsilon m\pi}{k_c^2 a}B\cos\dfrac{m\pi x}{a}\sin\dfrac{n\pi y}{b}e^{-j\beta z}$
Z	$Z_{\text{TE}}=\dfrac{k\eta}{\beta}$	$Z_{\text{TM}}=\dfrac{\beta\eta}{k}$

Note, $E_x = E_{0x}e^{-j\beta z}$

↳ Used only for simplifying derivations

→ Poynting Theorem (Law of conservation of EM energy)

Rate of decrease of EM energy in a volume = Power Flowing out through Surface + Power Dissipated as losses

We know, $\nabla.(E\times H) = H(\nabla\times E) - E(\nabla\times H)$

Take H conjugate,

$$\nabla.(E\times H^*) = H^*(\nabla\times E) - E(\nabla\times H^*)$$

Substitute Maxwell eq^{n}s

$$\nabla.(E\times H^*) = H^*(-j\omega\mu H - M_s) - E(J^* - j\omega\varepsilon^* E^*)$$
$$= -j\omega\mu|H|^2 - H^*M_s - J^*E + j\omega\varepsilon^*|E|^2$$
$$= -j\omega\mu|H|^2 - H^*M_s - J_s^*E - \sigma|E|^2 + j\omega\varepsilon^*|E|^2 \qquad (J^* = J_s^* + \sigma E^*)$$
$$= -\sigma|E|^2 + j\omega(\varepsilon^*|E|^2 - \mu|H|^2) - (J_s^*E + H^*M_s)$$

Integrate over volume & use divergence theorem $\left(\iiint_V \nabla.A\,dV = \iint_S A.dS\right)$

$$\iiint_V \nabla.(E\times H^*)\,dV = \int_V -\sigma|E|^2 + \int_V j\omega(\varepsilon^*|E|^2 - \mu|H|^2) - \int_V (J_s^*E + H^*M_s)$$

$$\iint_S (E\times H^*)\,dS = \int_V -\sigma|E|^2 + \int_V j\omega(\varepsilon^*|E|^2 - \mu|H|^2) - \int_V (J_s^*E + H^*M_s)$$

$\varepsilon = \varepsilon' - j\varepsilon''$ & $\mu = \mu' - j\mu''$ (Because it is lossy environment)

So, $$\iint_S (E\times H^*)\,dS + \int_V (E.J_s^* + H^*M_s)\,dV = \int_V -\sigma|E|^2 dV + j\omega\int_V (\varepsilon^*|E|^2 - \mu|H|^2)dV$$

$$-\frac{1}{2}\int_V (E.J_s^* + H^*M_s)\,dV = \frac{1}{2}\oint_S (E\times H^*)\,dS + \int_V \frac{\sigma}{2}|E|^2 dV + \frac{\omega}{2}\int_V (\varepsilon''|E|^2 + \mu''|H|^2)dV + \frac{j\omega}{2}\int_V (\mu'|H|^2 - \varepsilon'|E|^2)dV$$

$$P_s = P_0 + P_2 + 2j\omega(W_M - W_E)$$

- P_0: Power flow out of closed surface S
- P_2: Avg Power dissipated
- $2j\omega(W_M - W_E)$: Net reactive energy stored in volume

$\left(W_H = \frac{1}{4}\mu|H|^2,\ W_E = \frac{1}{4}\varepsilon|E|^2\right)$

Power Transmission in Rectangular Waveguide

Dominant mode in TE is TE_{10} where $m=1$ & $n=0$

So, $E_x = 0$

$$E_y = \frac{-j\omega\mu a}{\pi}\cdot A_{10}\cdot\sin\left(\frac{\pi x}{a}\right)e^{-j\beta z}$$

$E_z = 0$

$$H_x = \frac{j\beta a}{\pi}\cdot A_{10}\cdot\sin\left(\frac{\pi x}{a}\right)\cdot e^{-j\beta z}$$

$H_y = 0$

$$H_z = A_{10}\cos\left(\frac{\pi x}{a}\right)e^{-j\beta z}$$

Power flow down the guide for TE_{10}, $P_{10} = \frac{1}{2}\,Re\int_{x=0}^{a}\int_{y=0}^{b}(E\times H^*)\,\hat{z}\,dy\,dx$ (Using poynting theorem)

$$= \frac{1}{2}\frac{\omega\mu a^2}{\pi^2}|A_{10}|^2\,Re(\beta)\cdot\int_{x=0}^{a}\int_{y=0}^{b}E.H^*\,dy\,dx$$

$$P_{10} = \frac{\omega\mu a^3 b}{4\pi^2}|A_{10}|^2\,Re(\beta)$$

Power Loss in Rectangular Waveguide

Attenuation due to dielectric loss / conductor loss.

$R_s = \sqrt{\frac{\omega\mu}{2\sigma}}$, Surface Resistance

$$P_2 = \frac{R_s}{2}\int_S |J_s|^2 dl = R_s|A_{10}|^2\left(b + \frac{a}{2} + \frac{\beta^2a^3}{2\pi^2}\right)$$

Attenuation due to conductor loss, $\alpha_c = \frac{P_2}{2P_{10}} = \frac{R_s|A_{10}|^2\left(b+\frac{a}{2}+\frac{\beta^2a^3}{2\pi^2}\right)\cdot 4\pi^2}{2\omega\mu a^3 b\,|A_{10}|^2\beta} = \frac{R_s}{a^3b\beta\omega\mu}\left(2b\pi^2 + a^3k^2\right)$ Np/m

↳ For TE_{10}

→ The Power handling capacity of an air filled rectangular waveguide is usually limited by voltage breakdown which occurs at a field strength of about $E_d = 3\times10^6$ V/m for room temperature air at sea level pressure.
↳ Air can only withstand a limited Electric field before it ionizes

→ In an air filled rectangular waveguide, the electric field varies as $E_y = E_0 \sin\left(\frac{\pi}{a}\right)x$ which has a maximum value of E_0 at $x = \frac{a}{2}$ (middle of guide), therefore maximum power capacity before breakdown,

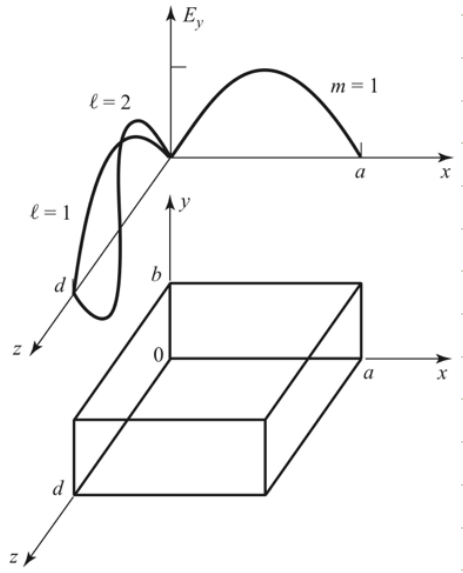
$$P_{max} = \frac{abE_0^2}{4Z_w} = \frac{abE_d^2}{4Z_w}$$

This shows power capacity increases with guide size $b = 2a$

Microwave Cavities

→ Closed metallic enclosure which store E & M energy, supports standing waves & resonates at discrete frequencies

→ Waveguides transport energy
Cavities trap energy & doesn't support travelling waves



→ Geometry (a, b, d) ⇒ width, Height & Length

In waveguide, $E_t(x,y,z) = \bar{e}(x,y)\left(A^+e^{-j\beta z} + A^-e^{j\beta z}\right)$

For cavity, at $z = 0$, $E_t = 0$ so $A^+ = -A^-$

& $E_t \propto \sin(\beta z) \Rightarrow \sin\beta z = 0$

$\beta z = p\pi$

$E_x = 0$

$E_y = E_0 \sin\left(\frac{\pi x}{a}\right)\sin\left(\frac{p\pi z}{d}\right)$

$E_z = 0$

$H_x = \frac{-jE_0}{Z_{TE}} \sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{p\pi z}{d}\right)$

$H_y = 0$

$H_z = \frac{j\pi E_0}{k\eta a} \cos\left(\frac{\pi x}{a}\right)\sin\left(\frac{p\pi z}{d}\right)$

Resonance Condition, $k_c = k_{mnp} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$

Resonant frequency, $f_r = \frac{1}{2\pi\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$

$TE_{m,n,p} \Rightarrow E_z = 0,\ H_z \neq 0$

$H_z = A_{mn} \cdot \cos\left(\frac{m\pi}{a}\right)x \cdot \cos\left(\frac{n\pi}{b}\right)y \cdot \sin\left(\frac{p\pi}{d}\right)z$

$TM_{m,n,p} \Rightarrow E_z \neq 0,\ H_z = 0$

$E_z = B_{mn} \cdot \sin\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y \cdot \cos\left(\frac{p\pi}{d}\right)z$

Vector Potential A for electric current source J

Take maxwell eq, $\nabla \times E = -j\omega\mu H$

$\nabla \times H = J + j\omega\varepsilon E$

Now we know, $\nabla \cdot B = 0$

So we define, $B = \nabla \times A$

but, $B = \mu H$

So, $H = \frac{\nabla \times A}{\mu}$

Now use, $\nabla \times H = J + j\omega\varepsilon E$

$\nabla \times \frac{\nabla \times A}{\mu} = J + j\omega\varepsilon E$

$\frac{1}{\mu}\left(\nabla(\nabla \cdot A) - \nabla^2 A\right) = J + j\omega\varepsilon E$

$= J + j\omega\varepsilon\left(-\nabla\phi_e - j\omega A\right)$ $\quad \left(E = -\nabla\phi_e - j\omega A\right)$

$= J - j\omega\varepsilon\nabla\phi_e + \omega^2\varepsilon A$

Then $-j\omega\varepsilon\nabla\phi_e - \frac{\nabla^2 A}{\mu} = J - j\omega\varepsilon\nabla\phi_e + \omega^2\varepsilon A$

$\left(\nabla \cdot A = -j\omega\mu\varepsilon\phi_e : \text{Lorenz condition}\right)$

$\hookrightarrow \nabla(\nabla \cdot A) = -j\omega\mu\varepsilon\nabla\phi_e$

$\hookrightarrow \phi_e = \frac{-\nabla \cdot A}{j\omega\mu\varepsilon}$ $\Rightarrow$ Lorentz Equation

$\omega^2\mu\varepsilon A + \nabla^2 A = -\mu J$

$k^2 A + \nabla^2 A = -\mu J$

Vector Potential F for magnetic current source M

Take Maxwell eq, $\nabla \times E = -M - j\omega\mu H$

We know, $\nabla \cdot D = 0$

Then we define, $D = -\nabla \times F$

But, $D = \varepsilon E$

So, $E = -\frac{\nabla \times F}{\varepsilon}$

Now, $\nabla \times E = -M - j\omega\mu H$

$\nabla \times \left(-\frac{\nabla \times F}{\varepsilon}\right) = -M - j\omega\mu H$

Similarly, $\nabla^2 F + k^2 F = -\varepsilon M$

$\phi_m = \frac{-\nabla \cdot F}{j\omega\mu\varepsilon}$

$H = -\nabla\phi_m - j\omega F$

	Electric Current	Magnetic Current
1)	$\nabla^2 A + k^2 A = -\mu J$	$\nabla^2 F + k^2 F = -\varepsilon M$
2)	$\phi_e = \frac{-\nabla \cdot A}{j\omega\mu\varepsilon}$	$\phi = \frac{-\nabla \cdot F}{j\omega\mu\varepsilon}$
3)	$E = -\nabla\phi_e - j\omega A$	$H = -\nabla\phi_m - j\omega F$
4)	$H = \frac{\nabla \times A}{\mu}$	$E = \frac{-\nabla \times F}{\varepsilon}$

Retarded Potential

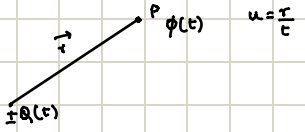
→ For the electromagnetic field generated by time varying electric current / charge distribution in past

→ Consider charge Q of varying magnitude wrt t in a free space considering P.D b/w point P & location of Q

→ Let distance of seperation be r, then $\vec{v} = \frac{\vec{r}}{t}$.

In free space, $\mu_0 = 4\pi \times 10^{-7}$ H/m

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m



Here P.D $\phi(t)$ is a function of t & effect of charge Q(t) will be experienced at point t after a delay $\Delta t = r/v$. The field at the field points lag their sources in an amount of propagation delay time Δt. There lagging field are called as retarded fields & time varying potentials produced by those retarded fields are called as retarded potentials.

$\phi = \pm\int \frac{\rho_v\, dv}{4\pi\varepsilon_0 r}$ $\qquad \phi(t) = \frac{\rho_v\left(t - \frac{r}{v}\right) dv}{4\pi\varepsilon_0 r}$

Inhomogeneous Wave Equation

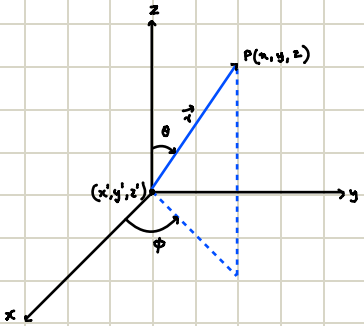
Inhomogeneous wave eqⁿ : $\nabla^2 A + \kappa^2 A = -\mu J$ ⇒ Inhomogeneous Helmholtz eqⁿ

homogeneous wave eqⁿ : $\nabla^2 A + \kappa^2 A = 0$

For Source at origin,

Assume current concentrated at origin, spherically symmetric solution, Only radial dependance

Let one component be $A_z = A_z(r)$



In spherical coordinates, $\nabla^2 A_z = \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dA_z}{dr}\right)$

$$= \frac{d^2 A_z}{dr^2} + \frac{2}{r}\frac{dA_z}{dr}$$

$$\Rightarrow \frac{d^2 A_z}{dr^2} + \frac{2}{r}\cdot\frac{dA_z}{dr} - \nabla^2 A_z(r) = 0 \longrightarrow (1)$$

Solving (1), we get 2 independant solutions,

$$A_{z_1} = C_1 \frac{e^{-jkr}}{r} \longrightarrow (2) \quad \text{(Outside Radiating Wave)}$$

$$A_{z_2} = C_2 \frac{e^{jkr}}{r} \longrightarrow (3) \quad \text{(Inside Radiating Wave)}$$

(2) & (3) represents an outwardly travelling wave & inwardly travelling wave respectively.

In this case, the source is at the origin, radiated fields travelling outward therefore solution is

$$A_z = A_{z_1} \longrightarrow (4)$$

Now if $K = 0$, wave eqⁿ reduces to $\nabla^2 A = -\mu J$ ⟶ Poisson's Equation

Solⁿ of eqⁿ ⇒ $A(r) = \frac{\mu}{4\pi}\iiint_v \frac{J_z}{r}\,dv'$ ⟶ Electrostatic-type solution

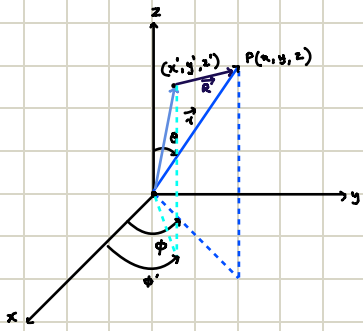
For dynamic/time-varying ⇒ $A_z = \frac{\mu}{4\pi}\iiint_v J_z\left(\frac{e^{-jkr}}{r}\right)dv'$

Similarly $A_x = \frac{\mu}{4\pi}\iiint_v J_x\left(\frac{e^{-jkr}}{r}\right)dv'$

$$A_y = \frac{\mu}{4\pi}\iiint_v J_y\left(\frac{e^{-jkr}}{r}\right)dv'$$

So the final solution $A = \frac{\mu}{4\pi}\iiint_v J\left(\frac{e^{-jkr}}{r}\right)dv'$

Now if source not at origin, $A = \frac{\mu}{4\pi}\iiint_v J\cdot\frac{e^{-jkr}}{r}\,dv' = \frac{\mu}{4\pi}\iiint_v J(x',y',z')\cdot\frac{e^{-jkr}}{r}\,dv'$



Similarly $F(x,y,z) = \frac{\varepsilon}{4\pi}\iiint_v M(x',y',z')\cdot\frac{e^{-jkR}}{R}\,dv'$

$$A = \frac{\mu}{4\pi}\iint J_s\cdot\frac{e^{-jkR}}{R}\,ds' = \frac{\mu}{4\pi}\int I_e\cdot\frac{e^{-jkR}}{R}\,dl'$$

$$F = \frac{\varepsilon}{4\pi}\iint_s M_s\cdot\frac{e^{-jkR}}{R}\,ds' = \frac{\mu}{4\pi}\int I_m\cdot\frac{e^{-jkR}}{R}\,dl'$$